Project Proposal Document – Capstone Project

Valentina Bernal Moreno – ARTSPACE

PROBLEM JUSTIFICATION

Artists lack an accessible platform to present their work to engaged audiences, while emerging artists and enthusiasts need a streamlined way to discover diverse visual art styles for inspiration.

BUSINESS JUSTIFICATION

- ArtSpace provides value by connecting artists with a wider audience and creating a community centered around discovering and sharing visual art. Artists can build a digital collection of their work and gain exposure, while users can discover artists and save network for inspiration.

TARGET USERS:

Art Enthusiasts:

- Users who enjoy discovering, browsing, and saving artwork they like
- Discover new artists and styles
- Browse galleries across many art forms

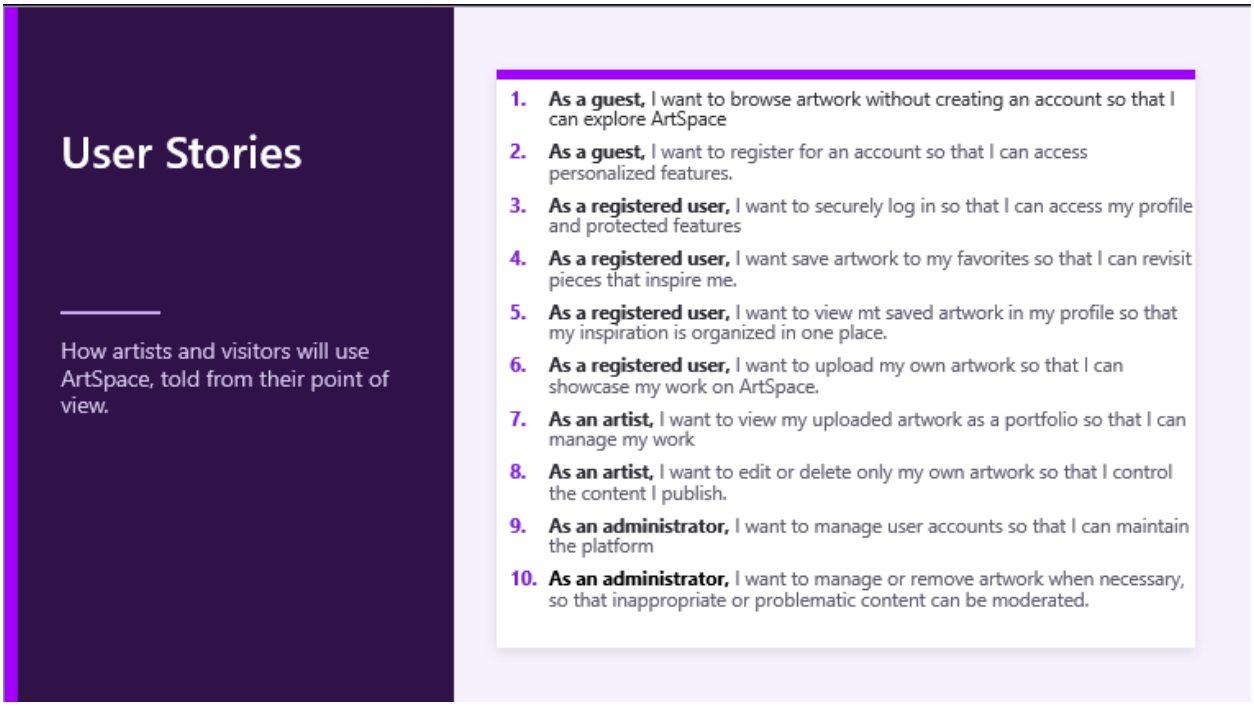
Registered Users:

- Users who want to upload, showcase and manage their artwork.
- Build a digital portfolio and gain exposure
- Reach people who appreciate their work

Admin Artists:

- Users learning painting, drawing, photography, etc. who want inspiration from other creators

- Study different styles and techniques
- Share progress and grow their skills



FUNCTIONAL REQUIREMENTS:

1. The system shall allow guests to browse artwork and categories without authentication.
2. The system shall allow new users to register for an account.
3. The system shall authenticate registered users and administrators.
4. The system shall allow authenticated users to add and remove artwork from their favorites.
5. The system allow authenticated users to view their saved favorites.
6. The system shall allow authenticated users to upload artwork with an image, title, description, and category.
7. The system shall provide users with a portfolio of their uploaded artwork.

8. The system shall allow users to edit and delete only artwork they own.
9. The system shall allow administrators to view and manage registered users.
10. The system shall allow administrators to manage artwork across the platform.
11. The system shall allow administrators to manage artwork categories.
12. The system shall restrict protected functionality based on authentication, role, and artwork ownership

NON-FUNCTION REQUIREMENTS:

Performance:

- The application should load artwork and user data efficiently without unnecessary delays
- API requests should return responses within a reasonable amount of time under normal usage.

Security:

- Users must be authenticated before accessing protected features such as uploading artwork and managing favorites.
- Users should only be able to edit or delete artwork that they uploaded.

Scalability:

- The application will use layered architecture to separate controllers, services, repositories, and data models
- The database should support growth in users, artwork, categories, and favorites.

OUT OF SCOPE ITEMS:

Commerce:

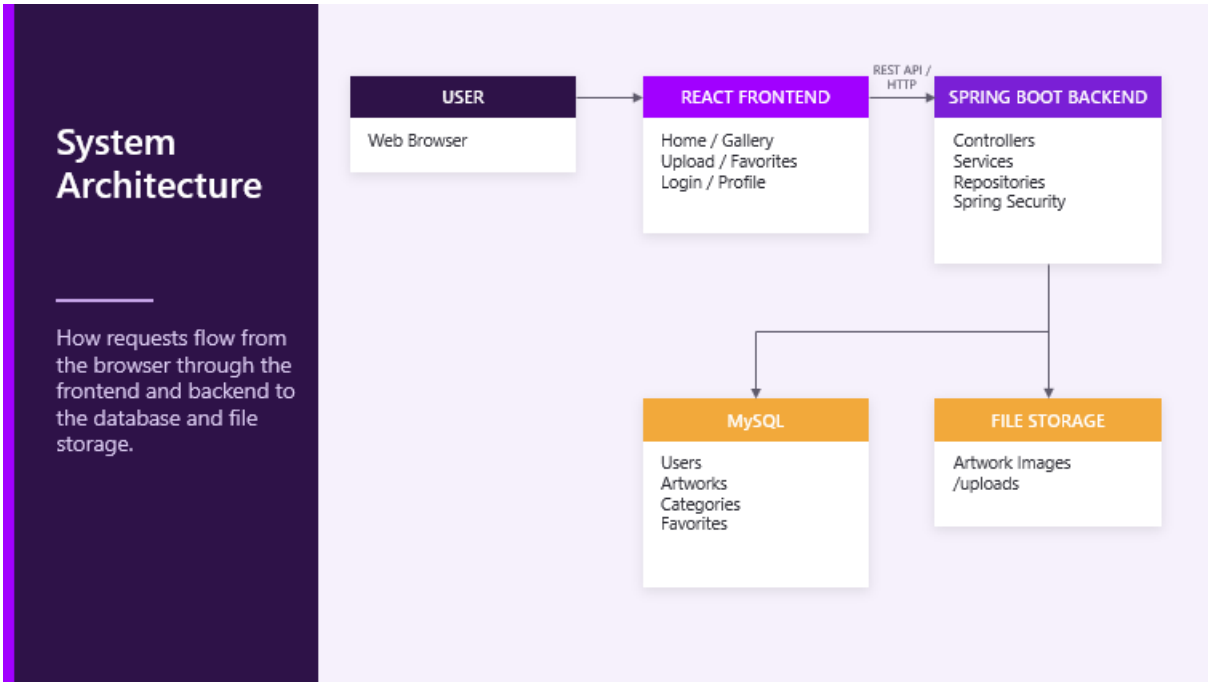
- Selling or purchasing artwork through the platform.
- Payment processing and financial transactions.

Social:

- Comments, likes, or ratings on artwork.
- Direct messaging between users.
- Following artists and activity notifications.

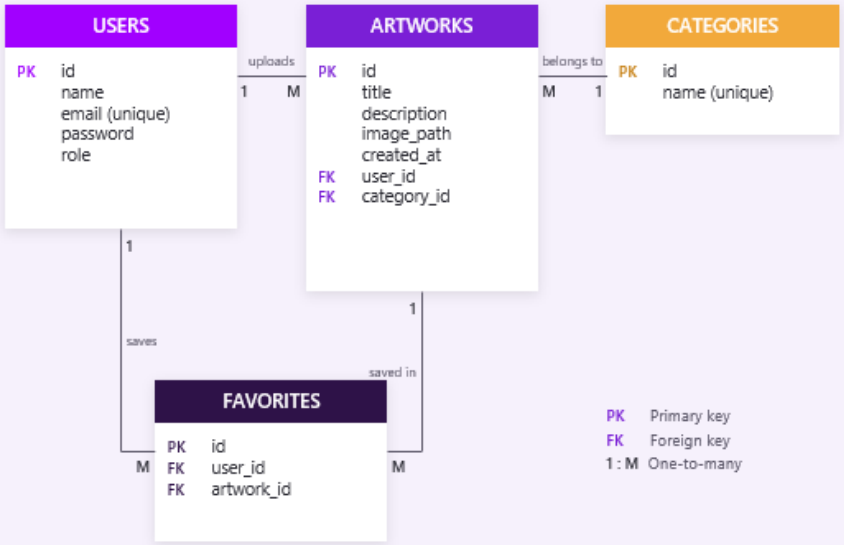
Platform:

- Native iOS and Android mobile apps.
- Admin dashboard for content moderation.



Entity Relationship Diagram

How ArtSpace stores users, artwork, categories, and favorites, and how they connect.



API Design

REST endpoints the React frontend uses to communicate with the Spring Boot backend.

Method	Endpoint	Purpose
POST	`/api/auth/register`	Register a new user
POST	`/api/auth/login`	Log in a user
GET	`/api/users/{id}`	Get user/profile information
GET	`/api/artworks`	View all artwork
GET	`/api/artworks/{id}`	View one artwork
GET	`/api/artworks/category/{categoryId}`	View artwork by category
GET	`/api/artworks/user/{userId}`	View artwork uploaded by a user
POST	`/api/artworks`	Upload/create artwork
PUT	`/api/artworks/{id}`	Edit an artwork
DELETE	`/api/artworks/{id}`	Delete an artwork
GET	`/api/categories`	View all categories
GET	`/api/favorites`	View logged-in user's favorites
POST	`/api/favorites/{artworkId}`	Add artwork to favorites
DELETE	`/api/favorites/{artworkId}`	Remove artwork from favorites

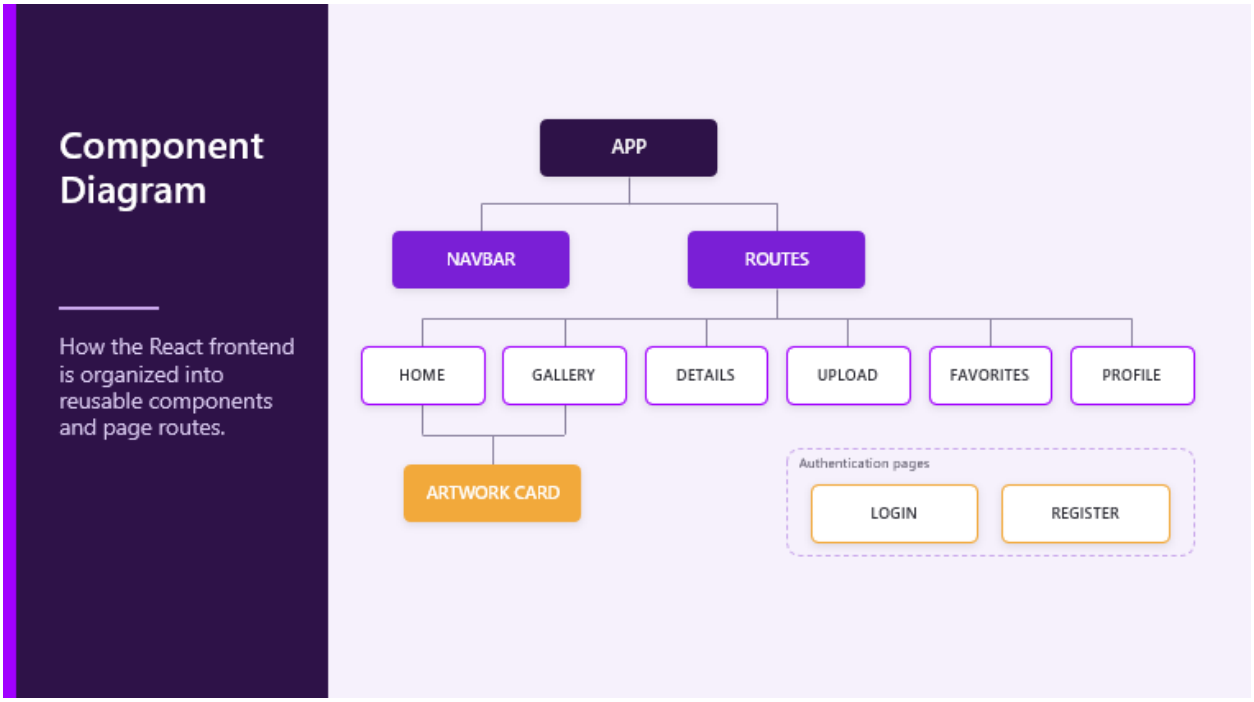
API DOCUEMNTATION:

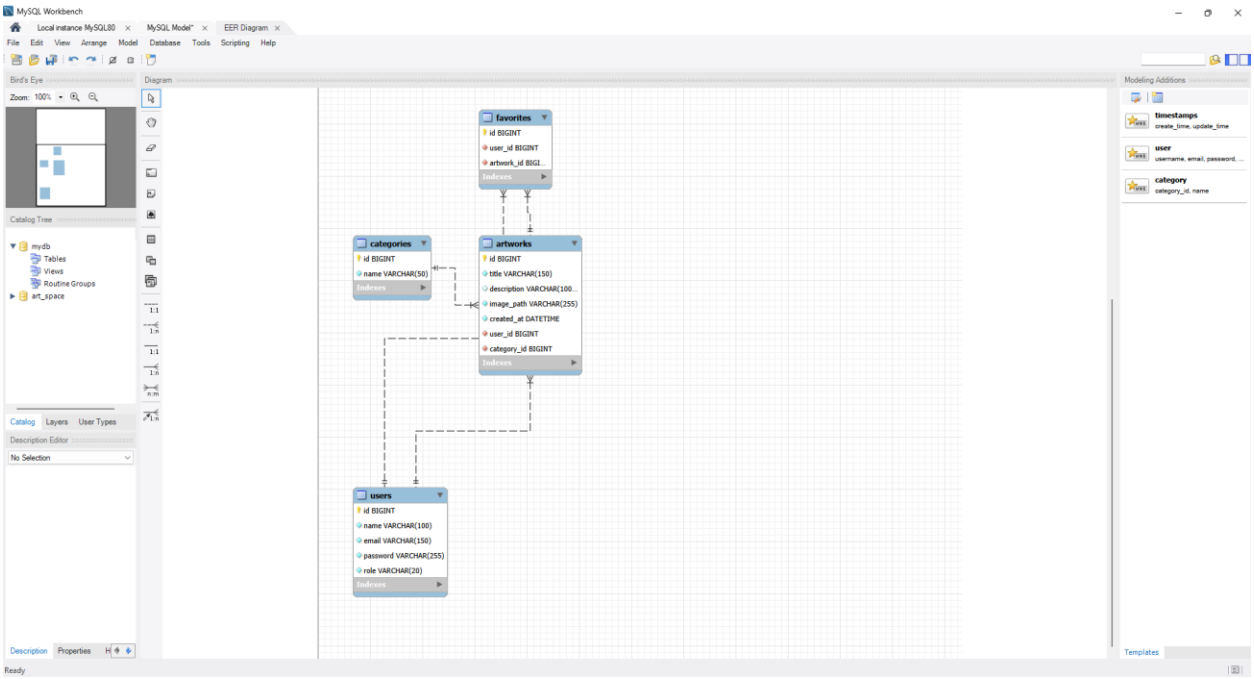
Pixabay API

ArtSpace integrates the Pixabay API to display additional artwork and photography on the Explore page.

- API Provider: Pixabay
- Official Documentation: https://pixabay.com/api/docs/
- Purpose: Retrieve images for users to discover and explore.
- Integration: The Spring Boot backend communicates with Pixabay and provides the results to the React frontend.

DATABASE DOCUMENTATION WITH ERD





Architecture Decision Records

1. Using MySQL for ArtSpace

Context: ArtSpace needs to store users, artworks, categories, and favorites with relationships between them.

Decision: Use MySQL as the relational database.

Reason: MySQL supports foreign keys, structured data, and integration with Spring Data JPA.

Trade-off: Database relationships require careful handling when deleting related records.

2. MySQL database: Store relational data and enforce relationships.

3. Spring Boot backend: Build REST APIs using Java and Spring Data JPA.

4. React frontend: Create a responsive, component-based interface.

5. Spring Security: Protect endpoints with authentication and role-based permissions.

6. Pixabay API: Add external artwork discovery without hosting every image yourself.

1. Deployment Guide — Step-by-Step Instructions

A Deployment Guide explains how someone else can set up and run your application.

Think of it as a tutorial for someone who downloads your ArtSpace project from GitHub and wants to get it working on their computer.

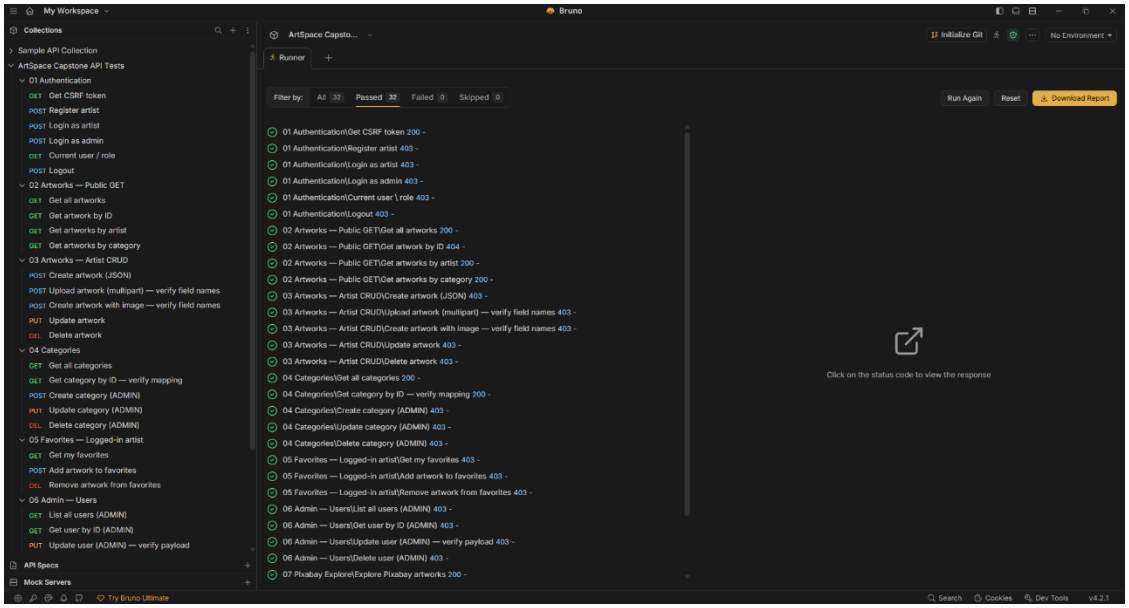
For ArtSpace, your deployment guide could include:

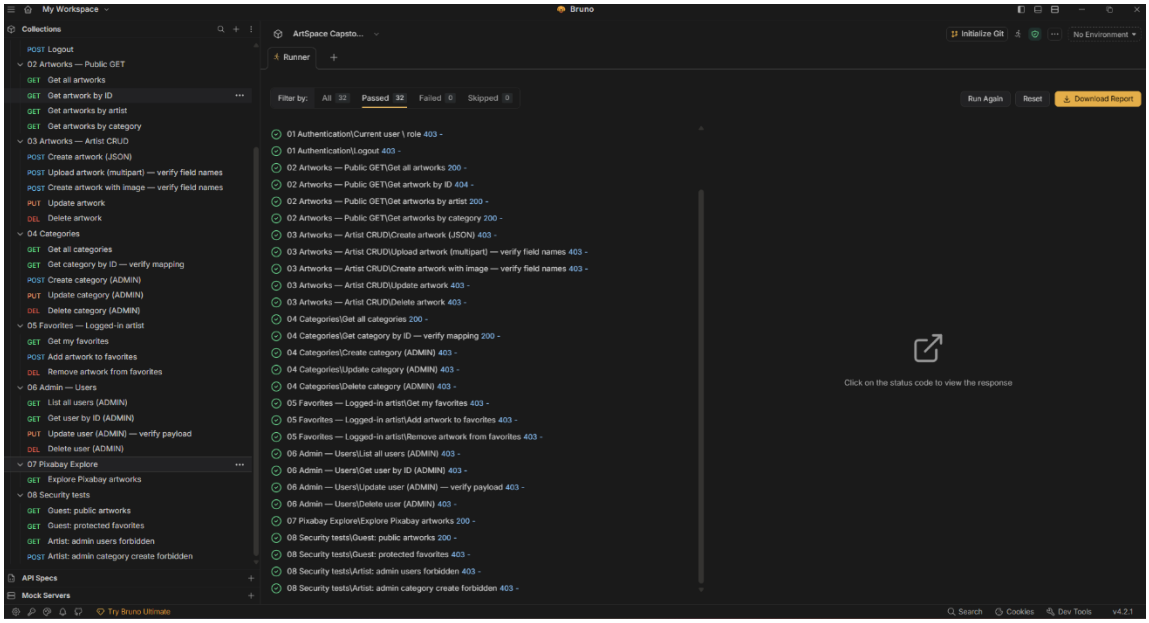
Step	Instructions
1. Prerequisites	Install Java 17, Node.js, MySQL, and Maven (or use the Maven wrapper).
2. Clone repository	Download the ArtSpace project from GitHub.
3. Database setup	Create the art_space database in MySQL.
4. Environment variables	Configure MySQL credentials and the Pixabay API key.
5. Start backend	Run the Spring Boot application in IntelliJ or through Maven.
6. Start frontend	Run npm install and npm run dev in the React frontend folder.
7. Test application	Open http://localhost:5173 and verify that the backend is running on port 8080.

Important: Since ArtSpace currently runs locally, you can write a Local Deployment Guide. You don't need to deploy it to AWS or another cloud provider unless your instructor specifically requires cloud deployment.

EXTRA DOCUMENTATION:

Postman / Bruno collection





Sonar Qube Report - Testing

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Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.example.artspace.controllers		52%		56%	24	58	74	169	13	33	0	6
com.example.artspace.services		84%		80%	9	55	18	125	3	29	0	5
com.example.artspace.models		72%		n/a	9	43	22	81	9	43	0	4
com.example.artspace.security		82%		n/a	3	12	9	34	3	12	0	1
com.example.artspace		37%		n/a	1	2	2	3	1	2	0	1
com.example.artspace.config		100%		n/a	0	3	0	10	0	3	0	1
com.example.artspace.exceptions		100%		n/a	0	6	0	10	0	6	0	1
Total	457 of 1,602	71%	32 of 102	68%	46	179	125	432	29	128	0	19

